

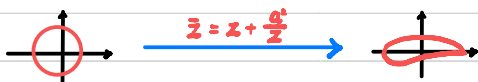
11/19/2023 Measurement of the pressure distribution on a circular cylinder Pre-lab

Introduction

Aiming · see the effect that **transition** and **separation** have on the pressure — **effect of viscosity**

· work out **lift** and **drag** on the body · show the relation between **Re** and **Co**

→ understand and predict behaviour of flow around more complex shape

e.g. Joukowski transformation 

Theory

Potential flow theory inviscid and incompressible

$$F(z) = U_{\infty} \left(z + \frac{R^2}{z} \right) - \frac{i\Gamma}{2\pi} \ln z = \phi + i\psi$$

* uniform stream + doublet + circulation

$$\therefore F(z) = U_{\infty} (re^{i\theta} + \frac{R^2}{r}e^{-i\theta}) - \frac{i\Gamma}{2\pi} \ln re^{i\theta}$$

$$= U_{\infty} [r(\cos\theta + i\sin\theta) + \frac{R^2}{r}(\cos\theta - i\sin\theta)] - \frac{i\Gamma}{2\pi} (\ln r + i\theta) = U_{\infty} (r\cos\theta + \frac{R^2}{r}\cos\theta) + iU_{\infty} (r\sin\theta - \frac{R^2}{r}\sin\theta) + \frac{\Gamma}{2\pi}\theta - i\frac{\Gamma}{2\pi}\ln r$$

$$\therefore \phi = U_{\infty} \cos\theta \left(r + \frac{R^2}{r} \right) + \frac{\Gamma}{2\pi}\theta \quad \text{and} \quad \psi = U_{\infty} \sin\theta \left(r - \frac{R^2}{r} \right) - \frac{\Gamma}{2\pi}\ln r$$

$$\therefore V_r = \frac{d\phi}{dr} = U_{\infty} \cos\theta \left(1 - \frac{R^2}{r^2} \right) \quad \text{and} \quad V_{\theta} = -\frac{d\psi}{dr} = -U_{\infty} \sin\theta \left(1 + \frac{R^2}{r^2} \right) + \frac{\Gamma}{2\pi r}$$

At surface of circular cylinder $r = R \quad \therefore V_r = 0 \quad \text{and} \quad V_{\theta} = -2U_{\infty} \sin\theta + \frac{\Gamma}{2\pi R}$

At stagnation point $V_{\theta} = 0 \quad \therefore \sin\theta = \frac{\Gamma}{4\pi R U_{\infty}}$

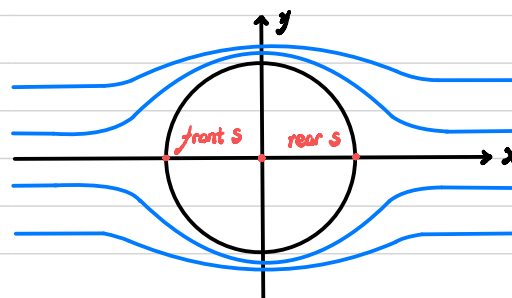
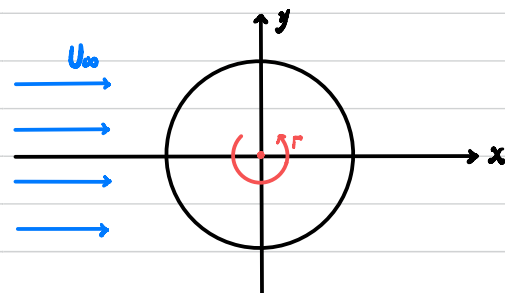
for no circulation $\Gamma = 0 \quad \therefore \sin\theta = 0 \quad \therefore \theta = 0 \text{ or } 2\pi$

Pressure coefficient $C_p = \frac{p - p_{\infty}}{\frac{1}{2}\rho U_{\infty}^2}$

Bernoulli's equation $p_0 = p_s + \frac{1}{2}\rho U^2$

$$\rightarrow p_0 = p_{\infty} + \frac{1}{2}\rho U_{\infty}^2 = p + \frac{1}{2}\rho U^2 \rightarrow C_p = 1 - \left(\frac{U}{U_{\infty}} \right)^2 = 1 - 4\sin^2\theta$$

cylinder is symmetric $\rightarrow D = 2 \int_0^{\pi} p_r \cos\theta d\theta \rightarrow C_D = \int_0^{\pi} C_p \cos\theta d\theta = 0 \quad \text{and} \quad C_L = \frac{1}{2} \int_0^{2\pi} C_p \sin\theta d\theta = 0$



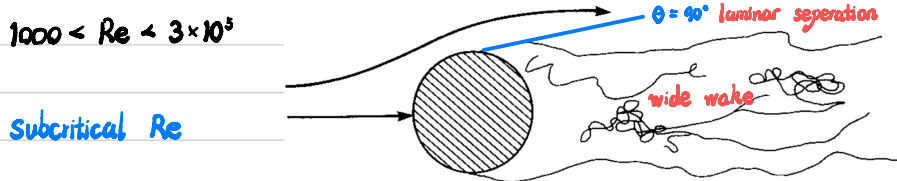
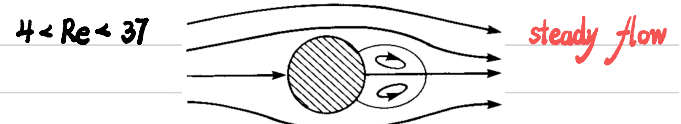
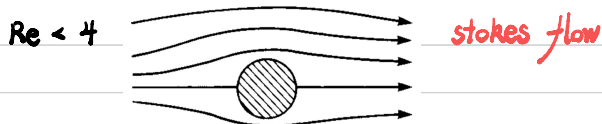
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Blockage effects considering the effect by wall interference, according to Rashko

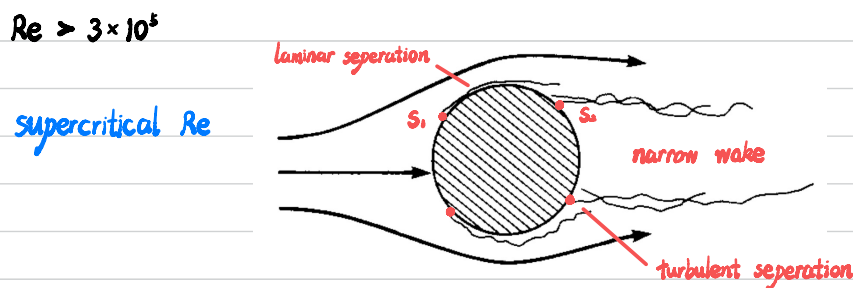
$$\frac{U}{U_{\text{measured}}} = 1 + \frac{C_d}{C_{d, \text{measured}}} \left(\frac{d}{h}\right) + 0.82 \left(\frac{d}{h}\right)^2 \quad \text{and} \quad \frac{C_d}{C_{d, \text{measured}}} = 1 - \frac{1}{2} C_{d, \text{measured}} \left(\frac{d}{h}\right) - 2.5 \left(\frac{d}{h}\right)^2 \quad \text{where} \quad \frac{d}{h} = \frac{\text{cylinder diameter}}{\text{tunnel breadth}}$$

$$\rightarrow (C_p - 1) = \left(\frac{U}{U'}\right)^2 (C_p' - 1)$$

Circular cylinder in realistic $Re = \frac{\rho U D}{\mu}$



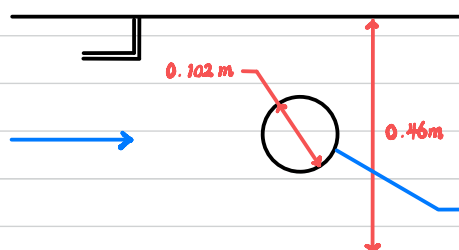
* critical $Re = 3 \times 10^5$



Procedure

Caution open the door of the room prevent blow things away downstream

Apparatus



$\rho = 789 \text{ kg/m}^3$ help to align the cylinder with 35 (p.)

Hole No.	1	2	3	4	5	6	7	8	9	10
θ (degrees)	-30	0	5	10	15	20	30	40	50	55
Hole No.	11	12	13	14	15	16	17	18	19	20
θ (degrees)	60	65	70	75	80	85	90	95	100	105
Hole No.	21	22	23	24	25	26	27	28	-	-
θ (degrees)	110	115	120	130	140	150	165	180	-	-

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Without a trip wire

① 15 ~ 16 m/s 700 rpm At 10°C, $Re = 1.066 \times 10^5$

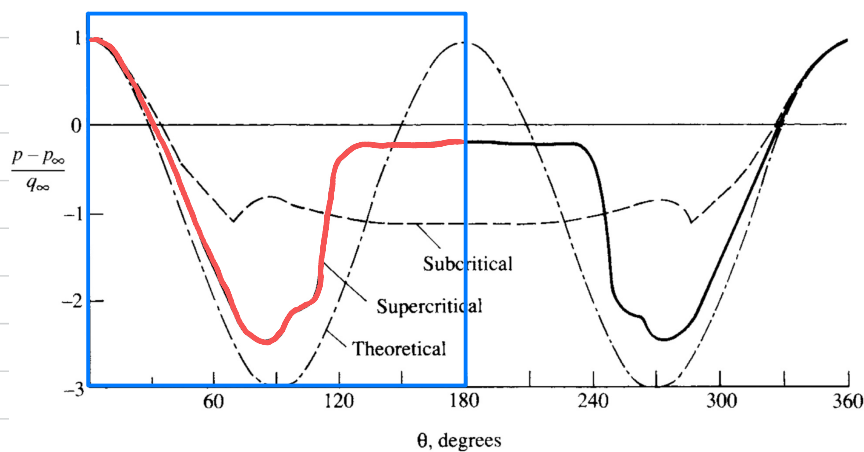
② 30 m/s 1400 rpm At 10°C, $Re = 2.132 \times 10^5$

With a trip wire

③ 15 ~ 16 m/s 700 rpm

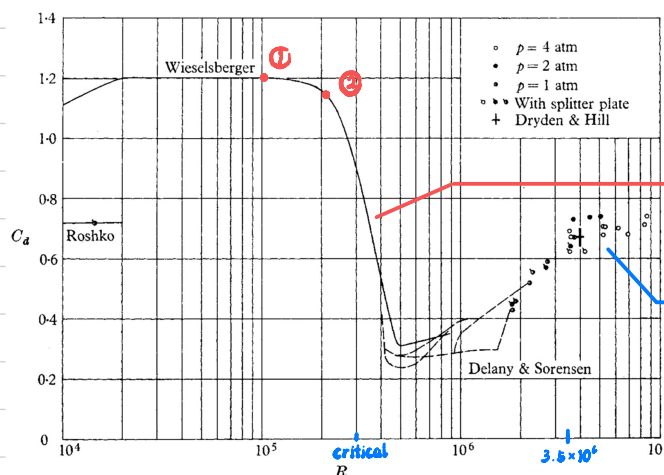
④ 30 m/s 1400 rpm

Prediction



Reference

Experiments on the flow past a circular cylinder at very high Reynolds number



corrections are important for large gradient

definite vortex shedding

* Change in C_d closely related to change in C_p back

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① As is well known, the (critical) Reynolds number at which the lower transition occurs can be lowered by introducing turbulence into the flow or roughness on the cylinder; we may expect a similar effect on the upper transition. the trip wire

② for wake shape

not important in this case

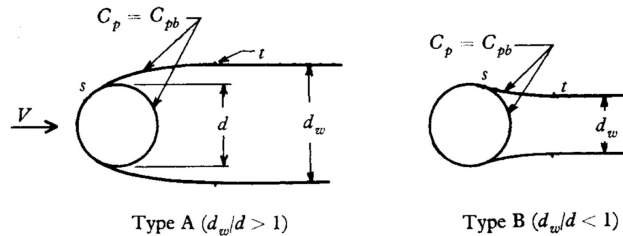


FIGURE 7. Free-streamline model for relating base pressure coefficient to wake width.

Now if the nature of the vortex shedding at transcritical Reynolds numbers is basically the same as at subcritical values, then the universal Strouhal number should apply to this range also. We corroborate this as follows. It can be shown from a simple momentum consideration (Eppler 1954) that, for flows of the type shown in figure 7,

$$C_d = (-C_{ps})(d_w/d). \quad (5)$$

③ conclusion

The existence of the upper transition makes it somewhat difficult to interpret the flow in terms of the classical description of the critical Reynolds number, which is roughly as follows. 'At subcritical Reynolds numbers the separation is laminar, and occurs early, on the front of the cylinder. With increasing Reynolds number, transition in the boundary layer moves ahead of the laminar separation point, the now turbulent boundary layer can withstand a greater pressure rise, and so separation moves to the rear of the cylinder, with a consequent decrease in the drag coefficient.' To include the new results in this picture we would now have to say that in the upper transition a turbulent separation moves forward, finally reaching a stationary point, still on the back of the cylinder, and C_d becomes constant.

It is probable that the actual events are somewhat more complicated. Bursnall & Loftin (1951) have pointed out that in the supercritical range there is evidence of a laminar separation 'bubble', that is, a localized region consisting of a laminar separation, transition, reattachment and turbulent separation. Such bubbles have been observed to play an important role in the stalling of thin airfoils. The occurrence of a bubble is indicated, for instance, by a 'flat' in a rising pressure distribution, such as that at about $\theta = 105^\circ$ in Flachsbarth's curve (figure 4). If the region of localized separation is small, the flat may not be so definite. It is possible that a separation bubble occurs throughout the supercritical range. On the other hand, it is certain that at sufficiently high Reynolds number there will be no bubble. We suggest that the disappearance of a separation bubble characterizes the transition from the supercritical to the transcritical régime. (We are indebted to Dr I. Tani for a discussion of this point.) This suggestion is tentative, subject to further experimental investigation, but we have no other explanation of the upper transition.

Thus we would propose the following classification: at subcritical Reynolds numbers the separation is laminar, in the supercritical range there is a laminar separation bubble followed by turbulent separation, and in the transcritical range the separation is purely turbulent.

experiment ① and ②

adding wire? not sure

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Procedure

Without a trip wire

① 15 ~ 16 m/s 700 rpm $T = 19.15^{\circ}\text{C}$

② 30 m/s 1400 rpm $T = 19.14^{\circ}\text{C}$

With a trip wire

③ 15 ~ 16 m/s 700 rpm $T = 19.35^{\circ}\text{C}$

④ 30 m/s 1400 rpm $T = 19.15^{\circ}\text{C}$

Adding ⑤ 7.5 m/s 350 rpm $T = 20.09^{\circ}\text{C}$

Error Analysis

Temperature $\pm 1^{\circ}\text{C}$ temperature keep rising during ③ and ⑤ due to sunshine * not considered

Manometer angle $\pm 0.2^{\circ}$ * not considered

Without a trip wire

① 15 ~ 16 m/s type 1-17 insignificant type 18-28 $\pm 0.1\text{cm}$ reading error $\pm 0.2\text{cm}$

② 30 m/s type 1-17 $\pm 0.1\text{cm}$ type 18-28 $\pm 0.2\text{cm}$ reading error $\pm 0.2\text{cm}$

With a trip wire

③ 15 ~ 16 m/s type 1-28 insignificant reading error $\pm 0.2\text{cm}$

④ 30 m/s type 1-17 insignificant type 18-28 $\pm 0.1\text{cm}$ reading error $\pm 0.2\text{cm}$

Extra test with a trip wire

⑤ 7.5 m/s type 1-28 insignificant reading error $\pm 0.2\text{cm}$

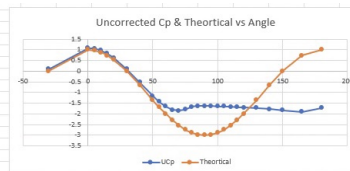
Data Recording

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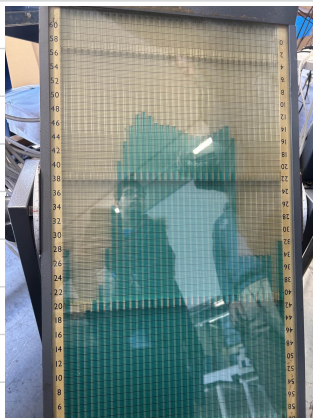
Excel sheet

palcochol (kg/m ³)
atm P (Pa)
T (°C)
T (K)
pair (kg/m ³)
b (Tunnel Breadth) (m)
d (Cylinder Diameter) (m)
Blockage Ratio
Dynamic Viscosity (Pa s)

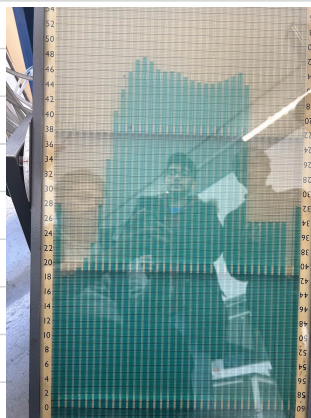
h total (cm) not running	32.2	50deg	Hole no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
19 15C		15 dm/s	θ (°)																												
h total (cm)	29.8		θ (rad)																												
h static (cm)	32.6	U = 15m/s	Tube height (cm)																												
C _p	0.9881		Δ height (m)																												
Re	0.9881		measured pres (Pa)																												
Separation Point		50deg	C _p																												
		30.2m/s	50																												
h total (cm)	17.8	1162rpm																													
h static (cm)	28	U = 30 m/s	Tube height (cm)																												
			Δ height (m)																												
		50deg	measured pres (Pa)																												
		15m/s	C _p																												
		506rpm																													
experiment 3	deg	Tripped	Tube height (cm)																												
total	30	U = 15 m/s	Δ height (m)																												
static	33		measured pres (Pa)																												
		50deg	C _p																												
		30m/s																													
		1305rpm																													
experiment 4	50 deg	Tripped	Tube height (cm)																												
total	17.6	U = 30 m/s	Δ height (m)																												
static	28.4		measured pres (Pa)																												
		Corrected U= 15m/s	C _p																												
		Corrected U= 30m/s																													
		Corrected, tripped, U=15																													
		Corrected, tripped, U=30																													
		manometer angle																													
	9.4	35 350rpm																													
	24.2	36 7.5m/s																													
20 1C		6deg	Tube height (cm)	13.4	8.2	8.4	8.8	9.6	10.4	12.8	16.8	19.8	20.2	21.4	22.2	22.4	22.4	21.8	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4
		Tripped 7.5m/s	Δ height (m)																												



Photos



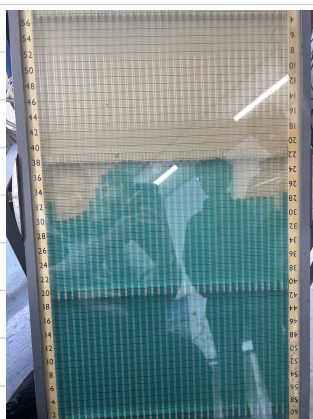
test 1



test 2



test 3



test 4



test 5

Data Analysis

— difference

Fig. 1: Theoretical C_p v. Angle

Fig. 2: Uncorrected C_p at different configurations v. Angle

corrected C_p is obviously larger

Fig. 3: Uncorrected and corrected C_p at different configurations v. Angle

separation points

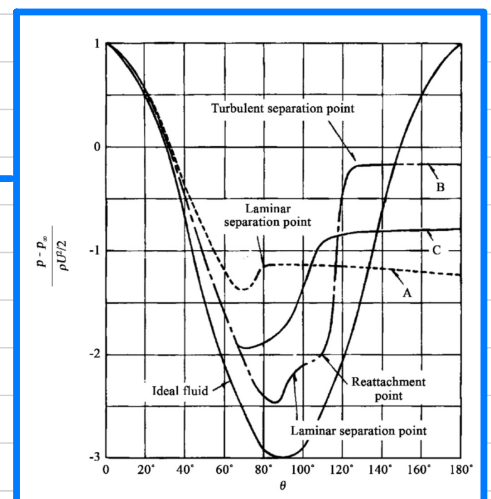


Fig. 4: Corrected C_p at different configurations v. Angle

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Test 5 was deemed an anomaly due to greater percentage error on the manometers at lower velocity

Discussion

Re

- $Re = \frac{\rho u D}{\mu}$ varies u in this case
- different flow regimes are characterized by different Re

Wind speed

- high wind speed results less relative error due to fluctuations
- higher U_{∞} gives greater Δh in manometer reduce relative uncertainty

Trip wire

- act as small circular cylinder induce turbulent as U cannot reach desired Re
- wake impinges on the boundary layer
- exact location is not essential, just placed in front of the separation
- ensure placed on symmetrical location for zero lift

cause flow separation $\longrightarrow$ undergoes transition then reattaches as a turbulent boundary layer

$\longrightarrow$ induce late separation * depend on Re $\longrightarrow$ decrease pressure drag

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Examples

- Golf balls delay point of separation by dimples
- Aircrafts turbulent boundary layer has better adhesion surface
- + dimples + free stream disturbances + dimples + periodic blowing or suction

Blockage effect

